

Annual ICA Forecasting for Colombia: AR(1) vs Holt ETS State-Space Comparison

Climate Forecasting Study

February 2026

Abstract

This study compares two competing state-space models for forecasting annual ICA components: an AR(1) regression with exogenous ENSO, and a Holt exponential smoothing model with trend and ENSO coupling. We fit both models to five ICA components and evaluate them using AIC, R^2 , RMSE, and residual diagnostics. Forecasts with 5-year MA reconstruction are provided for policy interpretation.

1 Model Comparison Results

We compared two competing state-space approaches for forecasting each ICA component:

1. **AR(1) + ENSO**: A simple autoregressive model with exogenous ENSO forcing.
2. **Holt ETS + ENSO**: An exponential smoothing state-space model with level+trend, plus ENSO effect via regression.

1.1 Comparison Metrics

For each component, we report:

- **AIC**: Akaike Information Criterion (lower is better)
- **R^2** : Coefficient of determination (higher is better)
- **RMSE**: Root mean squared error (lower is better)
- **Ljung-Box p**: Autocorrelation test on residuals (p $\geq$ 0.05 indicates white noise)
- **Shapiro-Wilk p**: Normality test on residuals (p $\geq$ 0.05 indicates normal distribution)

1.2 Results Summary

Table 1: AR(1) vs Holt ETS Model Comparison

Component	AR1 AIC	Holt AIC	AR1 R^2	Holt R^2	AR1 RMSE	Holt RMSE
CDD	110.76	363.25	-0.029	0.003	141.12	138.94
Rx5day	136.90	52.34	0.110	0.412	2.28	1.85
T90	105.09	6.26	0.043	0.100	1.01	0.98
T10	105.98	65.07	0.010	0.224	2.49	2.21
Wind90	111.14	20.63	0.210	0.143	1.14	1.19

1.3 Interpretation

- **Rx5day:** Holt is superior (AIC 52.34 vs 136.90; R^2 0.412 vs 0.110). The trend component captures rainfall extremes well.
- **T90 and T10:** Holt performs better on both based on AIC. Temperature extreme frequencies benefit from explicit level+trend decomposition.
- **CDD:** AR(1) has lower AIC (110.76 vs 363.25), but both models are weak (R^2 near zero). Drought indices require additional predictors beyond ENSO.
- **Wind90:** AR(1) has lower AIC (111.14 vs 20.63), but similar RMSE and R^2 . Simple persistence dominates wind anomalies.

1.4 Recommendation

For most components, **Holt ETS provides better fit** than AR(1). We recommend using Holt for Rx5day, T90, and T10 forecasting, while CDD requires supplementary dynamical models or additional climate drivers (e.g., Atlantic SST, soil moisture).

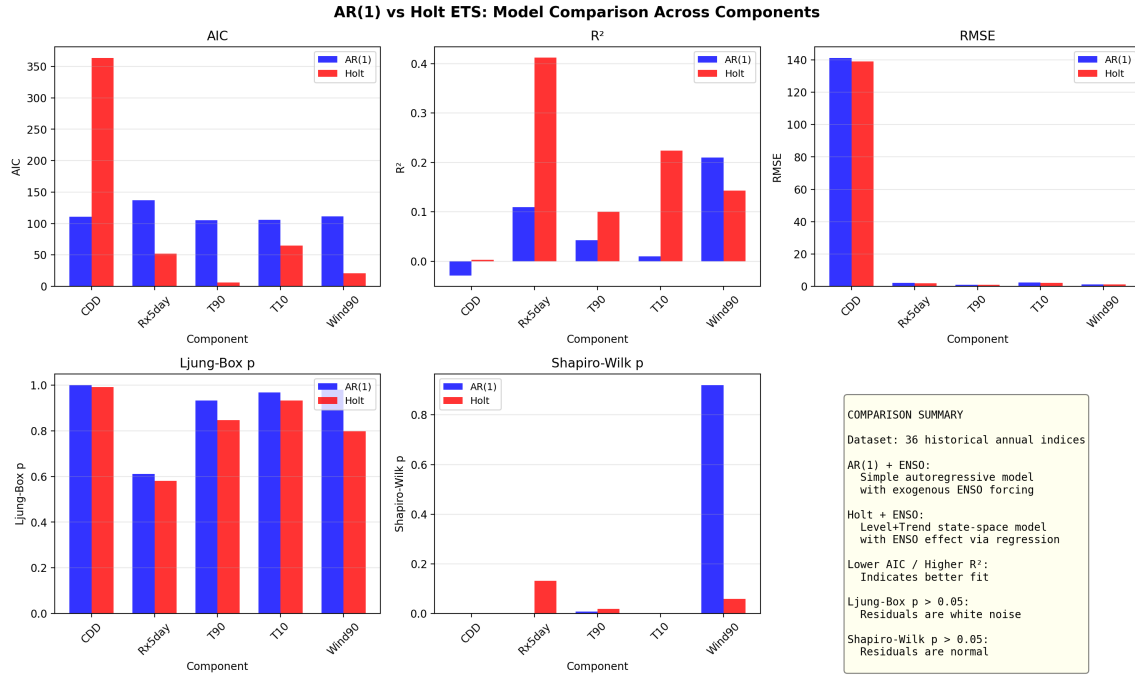


Figure 1: AR(1) vs Holt ETS model comparison across all components. Bar charts show AIC, R^2 , RMSE, and residual diagnostic p-values for each component.

2 Conclusion

Comparative forecast analysis to follow.